of light ray) (Figure 2.1).

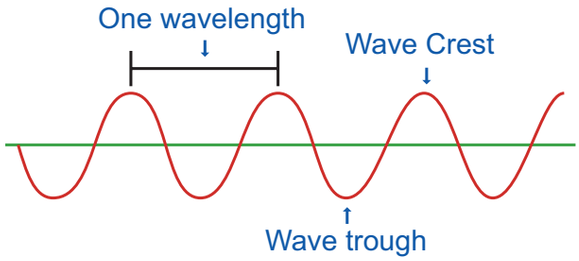



**Figure 2.1:** Wavelength-the distance between two adjacent crests or two adjacent troughs of the wave and denoted by greek letter (λ)

The sun produces a continuous spectrum of electromagnetic radiation with waves of various lengths (Figure 2.2). Radiation of longer wavelength includes Infrared (IR) and radiowaves, the shorter wavelengths include Ultra Violet (UV) rays and X-rays.

The physical behaviour of light can be caterigorised as either light rays, light waves or light particles. The combined properties of particle and wave enable light to interact with an object in several different ways like transmission, absorption, reflection, refraction, diffraction and scattering (Figure 2.3).

### 2.2.2 Lenses and its Properties

Lenses are optical devices which focus or disperse a light beam by means of refraction. A simple lens consists of a single piece of transparent material. Light rays from a distant source are focused at the focal point F. The focal point lies at a distance **f** (focal length) from the lens' centre (Figure 2.4).

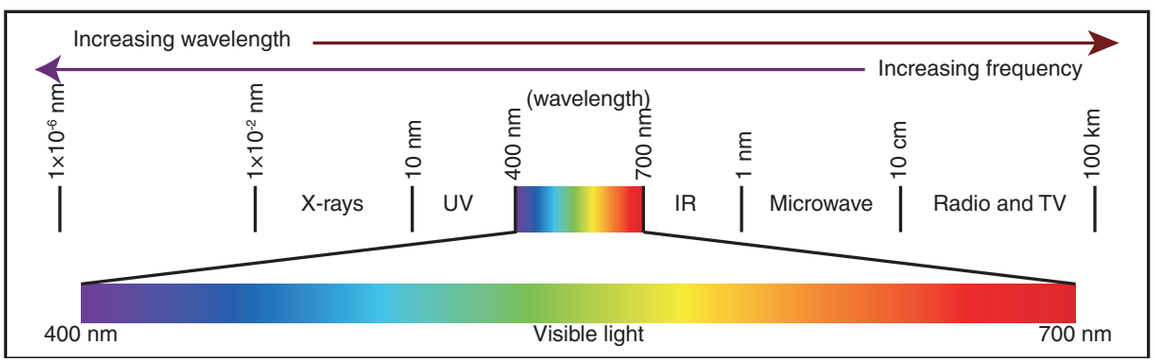



**Figure 2.2:** The electromagnetic spectrum-White light is a combination of all colours of visible spectrum

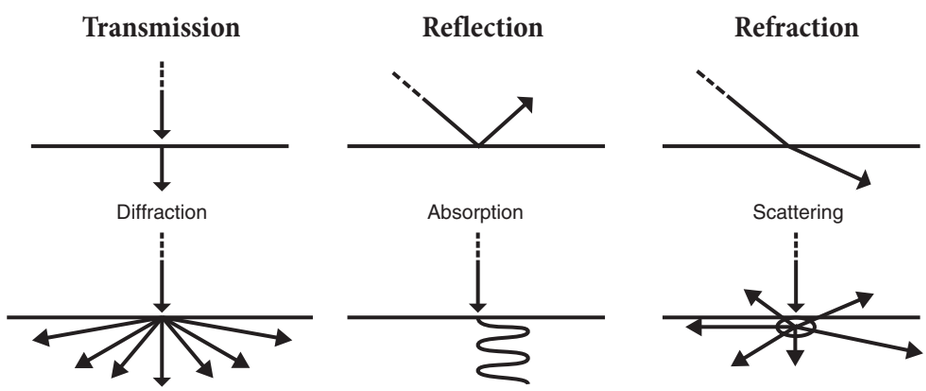



**Figure 2.3:** Interaction of light with matter



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